

Q3 - Difference between Link state and distance vector routing protocols

	Distance Vector	Link State
Knowledge of Topology	only knows about its immediate neighbors ("routing by rumor").	• Maintains a complete map (topology) of the entire network area.
Algo. used	uses Bellman-Ford algorithm.	• Uses Dijkstra (Shortest Path First algorithm).
Metric calculation	Primarily uses Hop count (number of routers).	• Uses cost, which is often based on bandwidth.
Update Method	Sends the entire routing table to neighbors.	• Sends link state Advertisements (LSAs) containing only link changes.
Update frequency	Periodic updates (sent at fixed intervals).	• Triggered updates (sent only when a change occurs).
Resource usage	Low CPU and memory requirements.	• High CPU and memory requirements to maintain the topology database.
Hierarchy	Flat network structure.	• Supports hierarchical structures (eg. OSPF Areas) to limit traffic.
Ex -	RIP (Routing Information Protocol), IGRP.	Ex - OSPF (Open Shortest Path First), IS-IS.